



DAY 5 • THE PROTOCOL

MAVLink

The protocol that connects the aircraft, the ground station, and your code.

~10 minutes

THE PROTOCOL

MAVLink — how everything talks

Every number on the GCS screen, every arm command, every waypoint — it is all MAVLink. The protocol that connects the entire drone ecosystem.

Dronecode — the open-source drone platform



PX4 / ARDUPILOT

The **firmware** running on the flight controller. We flashed ArduPilot — PX4 is the other major option. Both speak MAVLink.

MAVLink

The **protocol**. A compact binary format that defines how every component in the system talks to every other — FC to GCS, GCS to FC, companion computer to FC.

MAVSDK / QGC

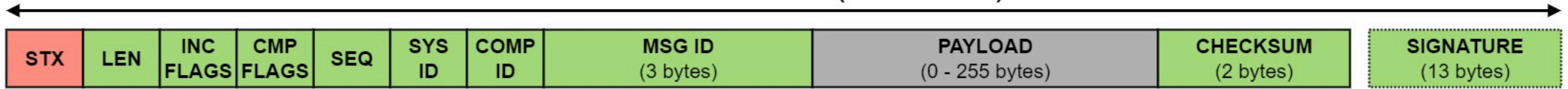
The **tools**. MAVSDK lets you write code that talks MAVLink. QGroundControl is a GCS built on it. Both are Dronecode projects.

MAVLink is the glue. **Learn the protocol and you can talk to any autopilot, from any language, on any platform.**

The MAVLink v2 packet

Every message between the aircraft and the ground is a binary packet with this structure.

MAVLink v2 Frame(12 - 280)



THE FIELDS THAT MATTER MOST

- SYS ID** Which vehicle (1 = first aircraft, 255 = GCS)
- COMP ID** Which component (1 = autopilot, 190 = companion)
- MSG ID** Which message — a 3-byte number, e.g. **#0 = HEARTBEAT**
- PAYLOAD** The actual data — attitude, position, command, etc.

WHAT THE GCS DOES WITH IT

The GCS receives hundreds of these per second. It reads the **MSG ID** to know what kind of data is in the payload, unpacks it, and draws it on screen. That is all a GCS is — a MAVLink decoder with a UI.

Messages vs Commands

MAVLink has two patterns — and the difference matters.

MESSAGES — STREAMING DATA

The FC sends these **continuously**, without being asked. Attitude, position, battery, GPS status. The GCS listens and displays them.

#0 HEARTBEAT — "I am alive, I am a copter, I am armed"

#33 GLOBAL_POSITION_INT — lat, lon, alt

#30 ATTITUDE — roll, pitch, yaw

#147 BATTERY_STATUS — voltage, current, remaining

COMMANDS — REQUESTS TO ACT

The GCS sends these **when you click something**. Arm, disarm, takeoff, go to waypoint, change mode. The FC replies with ACK or NACK.

#76 COMMAND_LONG — general purpose command carrier

400 MAV_CMD_COMPONENT_ARM_DISARM

22 MAV_CMD_NAV_TAKEOFF

16 MAV_CMD_NAV_WAYPOINT

Messages flow continuously, commands flow on demand. **If it is on the HUD, it is a message. If you clicked a button, it sent a command.**

Full message list: mavlink.io/en/messages/common.html